Theory Questions

a) Docker Fundamentals

- **Containers Vs VMs:** Containers uses the hosts kernel instead of loading a whole new OS with its dedicated kernel unlike VMs
- **Dockerfile:** set of instructions used to create a docker image
  `FROM`: must be first instruction and initializes the base image used
  `COPY`: copy files or dirs from host to container
  `RUN`: execute a shell command
  `ENTRYPOINT`: sets the default command when the container starts and can't be overridden
  `CMD`: sets a command to be executed when the container starts and can be overridden

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b) Image Management

- **Docker Images Layers:**
  - Each command in Dockerfile creates a new layer of executed changes relative to the previous layer
  - When building an image, docker caches layers and reuses it in later builds if the layer remains unchanged
- **Docker Volumes:**
  - Data persists if a container crashes or restarts as it is saved on the host machine and multiple containers can get mounted to the same volume to have access to the same data
  - Ex: When running a DB container(s), we want to persist data if the container(s) crash or restart and be able to allow other containers such as back-end containers to read data from the mounted volume to the DB container

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c) Networking in Docker

- **Network**: Docker has different types of networking:
  - **Bridge**: Creates a virtual network such that containers can communicate on the same bridge network via that network's internal IP addresses, and ports on host machines can be mapped to a specified container port

    - **Host**: the container uses the host machine network directly, such that if a container runs on port 8080, the host's port 8080 gets occupied by the container
    - **None**: Disables networking completely for the container
- **container-to-container network configuration**:
    1. create a network in docker
    2. attach both containers to the same network with the `--network` flag followed by the `<Network Name>`
    3. **(optional)**: specify names for each container to be able to directly communicate via service name instead of IP